PO11Q & PO12Q - Quantitative Political Analysis

Frequently Asked Questions

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Why should I take these modules?

- A vast amount of political research is quantitative. Even if you never conduct quantitative analysis yourself, a foundational understanding of these methods will allow you to **read, understand, and critically engage with the full breadth of scholarship in political science**. You have come to Warwick to study this discipline as a whole, and not just a subset that might seem more accessible.
- Your essays and assessments will require you to critically evaluate arguments, which means interrogating the evidence behind them. These modules give you the **tools to assess the strength of quantitative findings you cite or encounter in any piece of work you write**.
- In first year, graduation and entering the job market seem a long time off. But it pays to be prepared. Skills in data analysis are crucial for **finding employment in graduate level jobs**. In a 2021 policy paper the Department for Science, Innovation and Technology found that amongst the top 10 skills UK businesses wish applicants to have, three are directly related to data analysis: Analysis Skills, Database Management, and Data Processing, see Figure 1. Indeed, some civil service fast streams require evidence of formal statistical training.

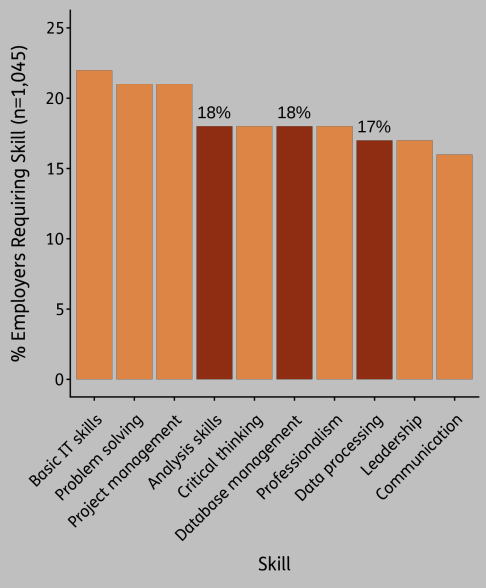


Figure 1: Top 10 skills required by UK businesses, source: Department for Science, Innovation and Technology, 2021

- PO11Q and PO12Q represent the first two modules of **a pathway in quantitative methods** in PAIS.

These modules form the prerequisites to the following modules:

- PO22Q: Introduction to Causal Inference in Quantitative Political Analysis
- PO31Q: Political Analysis with Maximum Likelihood Estimation
- QS305: Numbers in the Workplace. PAIS students who have successfully completed at least 30 CATs of quantitative methods modules (normally PO11Q and PO12Q) can apply for a place on this module. We have worked together with prestigious employers, such as the Office for National Statistics, YouGov, the European Social Survey, etc. with whom you will be matched for a four-week internship between second and third year of study. The module is credited in third year and is a great way to boost your CV!
- There are other, complementary modules in the curriculum, such as PO33Q (The Life and Death of Democracies and Dictatorships) which do not have PO11Q and PO12Q as prerequisites, but you will find these modules easier and more rewarding having laid the foundations before.

Who can take the modules?

- All first- and second-year students.
- If you are interested to learn and you are committed to weekly work (the module is strongly cumulative and is NOT organised into separate, self-contained topics like other modules in PAIS).

Who cannot take the modules?

- Third-year (finalist) PAIS students cannot take the module any more, as the university has changed its credit framework.
- If you want some easy modules to cruise through your first year.
- Note: in previous years, Politics and Sociology students could not take PO11Q in their second year, as Sociology's SO130 was an anti-requisite to PO11Q. This has been dropped, and the module is now open to Politics and Sociology and Sociology students, as well.

And what about maths? What if I need maths?

When you look at a jar of Nutella, there is a warning that it might contain nuts. And just as a hazelnut spread might contain nuts, quantitative methods modules might contain numbers and maths. But there is an important difference. Whilst nuts might be bad news for you if you are allergic, there is no such caveat for numbers and maths. The modules are designed to be accessible to students with no prior experience in quantitative methods, and all the required material will be taught on the module. I insist on teaching the modules in this way, because I am a strong believer that you should not just trust R (or ChatGPT for that matter) to give you the right answer. Very often, they do not, and you need to be sufficiently skilled to be able to spot when they are wrong. And this means to properly understand a method by knowing what is going on "under the hood".

But there is no need to panic, or to be put off. There is plenty of support available throughout the module (see "What support is available?" below). Indeed, several students have remarked in the past that they would have done a more quantitative degree, had they been taught the material in the way it is taught on these modules.

What is the module content?

- PO11Q: This module will deliver an introduction to quantitative political analysis. You will learn (1) how to turn abstract concepts, such as democracy, into a number, (2) how to describe numerical data, (3) how to infer from a small, representative sample to a much larger, never seen population with a quantifiable degree of certainty, (4) how to test theories empirically, and (5) the limits of such a process. The module uses the software R.
- PO12Q: We will start the module with an introduction to causality and correlation before moving to its main focus: regression analysis. This method is one of the standard tools of political science researchers and cannot only demonstrate whether relationships between variables exist, but also quantify the magnitude and direction of such a relationship. We will progress in small steps, starting with only one independent variable to explore the intuition of regression, its interpretation, and goodness of fit. We will then extend this to multiple independent variables, explore variable transformations, and finally the assumptions underpinning the method.
- For detailed syllabi, see Table 1.

Table 1: Syllabi – PO11Q and PO12Q

PO11Q	PO12Q
1. Appointment with Evidence: Using the Power of Numbers	1. Because... Reasons: Causal Thinking
2. How to turn a Concept into a Number	2. Detecting Patterns with Correlation
3. Conceptualisation and Measurement in Practice	3. Beyond Patterns with Linear Regression
4. Understanding Single Variables	4. Ordinary Least Squares: A Look under the Hood
5. Types and Shapes of Data	5. OLS: Why and How Well Does It Work?
6. (Reading Week)	6. (Reading Week)
7. Working with Variation: Probability and Sampling	7. From Sample to Population in Regression
8. Measuring Uncertainty: Thinking in Intervals	8. The Real World: Multiple Regression
9. From Sample to Population	9. Transformation of Variables: Unlocking Patterns
10. Why Statistics Can Mislead: Errors and Power Explained	10. The Fine Print: Making Your Regression BLUE

What software do the modules use?

- Both modules use the software R.

How are the modules taught?

- A 1-hour lecture per week
- A 2-hour seminar per week
- The seminars are taught through dedicated seminar companions (PO11Q and PO12Q, password: "Darwin20191021"), hosted on my GitHub page.

> **Important:**
> The content of these modules is strongly cumulative. The modules require sustained, weekly work in order to succeed. You will not be able to understand Week 8, for example, if you have not understood Week 7. Consistent attendance is expected as a vital component of the modules.

What support is available?

With the warning of cumulative content also comes a lot of support to enable you to do well on the modules. This includes, but is not limited to:

- Detailed lecture slides with full glossary and references
- Seminar companions (click PO11Q and PO12Q, the password is "Darwin20191021") with:
  - Further explanations
  - Explanatory videos / mini tutorials
  - Interactive applications, see here for an example to calculate descriptive statistics by hand
  - A weekly glossary for new terms (see here for the full glossary of PO11Q)
  - Weekly flashcards to learn R (click PO11Q and PO12Q to see examples)
  - Weekly exercises
  - Full solutions to all exercises
- A weekly, non-assessed quiz
- Discussions and questions in seminars
- Weekly office hours
- Email support
- Forums for peer discussion

When do the modules run?

- PO11Q runs in Term 1 (Autumn)
- PO12Q runs in Term 2 (Spring)
- NB: PO12Q is not PO11Q being run again at a later time. The two modules have distinct syllabi, see Table 1

How are the modules assessed?

- PO11Q is assessed through two components:
  - A 750-word report on the conceptualisation and measurement of a social science concept (choice of concepts provided), worth 25%
  - A 2-hour, written exam which will take place in person, on campus in January, worth 75%
- PO12Q is assessed through two components:
  - An in-class test which will take place in person in Week 7, worth 10%
  - A 2-hour, written exam which will take place in person, on campus in the summer examination period (Term 3), worth 90%

Can I only take PO11Q?

- Absolutely fine. PO11Q is a standalone module and does not require you to take PO12Q.
- What I will say, however, is that PO12Q is the cooler module, as we will be exploring the relationships of two or more variables with each other on that module. And this is the foundation for actually answering proper research questions.

Can I only take PO12Q?

- Only in exceptional circumstances. PO11Q is a required prerequisite for this module. If – and only if – you have done an International Baccalaureate (IB), you may apply to take PO12Q without taking PO11Q. You will need to provide evidence of your prior learning, such as a syllabus. If you are interested in this option, please contact me as soon as possible.

List of References

Department for Science, Innovation and Technology. (2021). *Quantifying the UK Data Skills Gap: Full Report*. https://www.gov.uk/government/publications/quantifying-the-uk-data-skills-gap/quantifying-the-uk-data-skills-gap-full-report#defining-data-skills